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**Supplementary Material**

# Supplementary Material: Clean Breaks: Tracking Recombination as Continuous Motion Across Sequence Manifolds

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## Abstract

**This document contains supplementary tables and supporting materials for the manuscript "Clean Breaks: Tracking Recombination as Continuous Motion Across Sequence Manifolds".**

**Keywords** Recombination, Reticulate Evolution, Simulation Benchmark, Supplementary Material

## Supplementary Material

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**Table S1 Quantitative comparison of `RhizAeon` and 3SEQ across the 3,200-trial Posada & Crandall (2001) benchmark suite.** Results are grouped by evolutionary regime: False Positive evaluation under Gamma rate heterogeneity ($\alpha \in \{2.0, 0.5, 0.05\}$, $\rho = 0$, 1,200 trials), Clock-like null controls ($\rho = 0$, 400 trials), and Recombinant coalescent trials ($\rho \in \{1, 4, 16, 64\}$, 1,600 trials). Columns report the population mutation parameter ($\theta$), population recombination parameter ($\rho$), alignment-level power / FPR (%), mean inferred event count ($\hat{K}$), precision (PPV), triad lineage attribution accuracy ($S_{\text{triad}}$), Informative SNP Jaccard spatial overlap ($J_{\text{SNP}}$), and execution runtime per alignment (ms).

| | | `RhizAeon` (Continuous Manifold Tracking) | | | | | 3SEQ (Discrete Triplet Scanning) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Evolutionary Regime | Trials | Power / FPR (%) | $\hat{K}$ | Precision | $S_{\text{triad}}$ | Time (ms) | Power / FPR (%) | $\hat{K}$ | Precision | $S_{\text{triad}}$ | Time (ms) |
| **A. Rate Heterogeneity Null Suite ($\rho = 0$, Empirical False Positive Calibration)** | | | | | | | | | | | |
| $\theta = 10, \alpha \in [0.05, 2.0]$ | 300 | **0.0%** | 0.00 | 100.0% | 1.00 | 19.0 | 0.0% | 0.00 | 100.0% | 1.00 | 28.2 |
| $\theta = 50, \alpha \in [0.05, 2.0]$ | 300 | **0.3%** | 0.00 | 99.7% | 1.00 | 22.0 | 0.0% | 0.00 | 100.0% | 1.00 | 26.6 |
| $\theta = 100, \alpha \in [0.05, 2.0]$ | 300 | **0.7%** | 0.01 | 99.3% | 0.99 | 20.9 | 0.3% | 0.00 | 99.7% | 1.00 | 27.4 |
| $\theta = 200, \alpha \in [0.05, 2.0]$ | 300 | **1.3%** | 0.01 | 98.7% | 0.99 | 24.3 | 0.3% | 0.00 | 99.7% | 1.00 | 29.2 |
| *All Rate-Variation Nulls* | *1,200* | **0.58%** | **0.01** | **99.4%** | **0.99** | **21.5** | *0.17%* | *0.00* | *99.8%* | *1.00* | *27.9* |
| **B. Recombinant Coalescent Suite ($\rho > 0$, Statistical Power and Lineage Attribution)** | | | | | | | | | | | |
| $\theta = 10$ (Low Divergence) | 400 | **11.8%** | 0.14 | 17.9% | 23.7% | 15.1 | 10.3% | 0.26 | 5.6% | 19.7% | 31.3 |
| $\rho = 1$ | 100 | 3.0% | 0.03 | 55.4% | 55.5% | 10.9 | 2.0% | 0.04 | 54.0% | 54.2% | 41.1 |
| $\rho = 4$ | 100 | 13.0% | 0.17 | 24.4% | 25.0% | 11.9 | 16.0% | 0.40 | 20.1% | 21.2% | 37.3 |
| $\rho = 16$ | 100 | 26.0% | 0.30 | 9.5% | 11.6% | 20.2 | 22.0% | 0.58 | 2.1% | 3.2% | 23.2 |
| $\rho = 64$ | 100 | 5.0% | 0.05 | 2.3% | 2.7% | 17.5 | 1.0% | 0.03 | 0.2% | 0.3% | 23.7 |
| $\theta = 50$ (Moderate Divergence) | 400 | **67.8%** | 1.51 | 40.7% | 54.2% | 19.8 | 83.5% | 6.97 | 16.6% | 34.6% | 27.6 |
| $\rho = 1$ | 100 | 33.0% | 0.44 | 55.2% | 58.0% | 15.6 | 38.0% | 2.14 | 41.0% | 48.5% | 31.7 |
| $\rho = 4$ | 100 | 57.0% | 0.90 | 34.3% | 37.6% | 17.7 | 73.0% | 5.18 | 7.0% | 17.0% | 28.9 |
| $\rho = 16$ | 100 | 88.0% | 2.09 | 34.3% | 45.2% | 25.8 | 96.0% | 9.75 | 6.4% | 19.1% | 24.5 |
| $\rho = 64$ | 100 | 93.0% | 2.62 | 39.1% | 58.1% | 20.2 | 100.0% | 10.82 | 11.9% | 33.8% | 25.4 |
| $\theta = 100$ (Substantial Divergence) | 400 | **73.5%** | 1.87 | 43.1% | 53.2% | 21.6 | 84.3% | 10.23 | 14.2% | 29.5% | 31.2 |
| $\rho = 1$ | 100 | 37.0% | 0.48 | 55.9% | 59.7% | 15.4 | 52.0% | 3.84 | 34.9% | 46.9% | 36.7 |
| $\rho = 4$ | 100 | 69.0% | 1.20 | 43.3% | 48.4% | 18.4 | 85.0% | 8.38 | 6.3% | 17.9% | 31.8 |
| $\rho = 16$ | 100 | 90.0% | 2.29 | 35.1% | 45.1% | 22.7 | 100.0% | 14.17 | 5.4% | 19.4% | 27.8 |
| $\rho = 64$ | 100 | 98.0% | 3.51 | 38.1% | 59.5% | 29.4 | 100.0% | 14.54 | 10.3% | 33.5% | 28.4 |
| $\theta = 200$ (High Divergence) | 400 | **80.0%** | 2.49 | 45.2% | 56.1% | 22.5 | 83.0% | 12.12 | 16.4% | 31.3% | 36.8 |
| $\rho = 1$ | 100 | 43.0% | 0.52 | 64.6% | 67.3% | 17.1 | 43.0% | 4.05 | 46.5% | 55.2% | 39.6 |
| $\rho = 4$ | 100 | 83.0% | 1.58 | 38.6% | 45.7% | 20.4 | 89.0% | 10.92 | 4.6% | 16.7% | 37.0 |
| $\rho = 16$ | 100 | 96.0% | 3.28 | 39.2% | 53.4% | 23.4 | 100.0% | 17.25 | 5.6% | 21.2% | 37.3 |
| $\rho = 64$ | 100 | 98.0% | 4.56 | 38.4% | 58.2% | 29.4 | 100.0% | 16.26 | 9.0% | 32.6% | 33.3 |